

POWERFACTORY

PowerFactory 2024

Technical Reference

Measurement Block

RelMeasure, TypMeasure

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The Measurement block is used to calculate the RMS values and sequence components of current and voltage. The block receives the inputs from a Ct and / or a Vt block. The function performed by a Measurement block will depend on whether a static or a dynamic simulation is being carried out. For the dynamic simulation it can be selected between analogue and digital measurement.

Depending on the combination of measurement selection, number of phases and kind of measurement, different calculation models are available. The following sections describe the models and combinations for each.

2 Measurement Type Input (*TypMeasure*)

2.1 Basic Data

The *No. Phases* (*e:iphases*) defines the number of output loops available. Current and voltage measurement can be enabled separately.

The nominal current and voltage ranges (*e:rInom* and *e:rUnom*) are mandatory if the corresponding measurement is enabled. The additional *Current Scaling* range (*e:rIs*) can be used to model LVCB rating plugs.

For 3- and 6-phase measurements, the zero sequence treatment can be defined to one of the following:

- *Measured*: The zero sequence will be taken from the corresponding zero sequence input signal;
- *Calculated*: The zero sequence will be calculated from the three phase input signals;
- *User-defined*: The selection can be made in the element (see Section 3).

The selection can be made separately for current (*e:icalI0*) and voltage (*e:icalU0*) measurement.

For a 3-phase measurement, an additional selection whether the output loops are “Phase-phase” or “Phase-earth” can be made. The selection can be made separately for current and voltage measurement.

2.2 EMT Simulation

For dynamic simulations an analogue measurement is selected by definition of the “Measuring time”. A digital measurement can be selected by activating the check box “RMS calculation”.

2.2.1 Analogue Measurement

A 2nd order analogue filter is used if the check box “RMS calculation” is not activated. The algorithm executed by the transfer procedure each simulation step is using two internal variables (S_1, S_2) and their derivative (S_1', S_2'):

The 1st and 2nd internal variable derivative calculated to

$$S_1'(n) = 2\pi f \Big(v(n) - v(n-1)\Big). \tag{1}$$

Where:

$v(n)$ is the input value read during the n step of the simulation.

f is the system frequency.

The 2nd internal variable derivative S_2' is calculated under consideration of the filter time constant T_f:

$$S_2'(n) = \frac{v(n) - S_1(n) - S_2(n)}{T_f}. \tag{2}$$

The filter output value v_{out} is calculated for every n step of the simulation:

$$v_{\text{out}}(n) = \frac{\sqrt{2}\, S_2(n)}{\sqrt{1 + w^2\, T_f^2}} \tag{3}$$

2.2.2 Digital Measurement

For the digital measurement, the RMS calculation method (*e:idftenabled*) determines which algorithm type is used to calculated the RMS values. For this purpose, the combo box shows four items: “Integral”, “FFT”, “DFT” and “Cosine”.

The number of calculation samples of the input signal given during each cycle is defined in the field “RMS calculation samples”. This value is used by the “Integral” and the “FFT” algorithms.

For the RMS calculation methods “DFT” and “Cosine”, the harmonic order (*e:dftharmonic*) can also be defined with a value > 0. Only the harmonic specified in the parameter is extracted from the input samples. The parameter last harmonic (*e:fftlastharm*) define the last harmonic which is extracted from the input current and voltage samples by the “FFT” algorithm. A value > 0 may be entered. The specified harmonic parameter is the higher harmonic extracted from the input samples.

Integral Method
When $v(t)$ is sampled, the resulting sample values are denoted by $v(n)$. Since $v(k)$ represents sampled values with a fixed sample rate of N samples per cycle of sinusoidal value. The “Integral” method calculates the RMS magnitude as follows.

$$V(n) = \sqrt{\frac{1}{N} \sum_{k=n-N+1}^{n} v(k)^2} \tag{4}$$

Where:

$V(n)$ is the RMS value.

N defines the number of RMS calculation samples per cycle (*e:meassamples*).

$v(k)$ is the input value of a series of samples.

The integral method can only calculate magnitudes and therefore cannot differentiate between real and imaginary parts. The calculated magnitude is assigned to the real part of the signal. The imaginary part is kept at zero. Therefore, a transformation of the output quantities using the symmetrical components transformation in positive- and negative-sequence value is not possible.

Discrete Fourier Transform

The sinusoidal form of a value $v(t)$ with the angular frequency $\omega = 2\pi f$ can be described with a cosine function:

$$v(t) = \hat{v}\ \cos(\omega t + \varphi_v)\,. \tag{5}$$

This can be expanded as:

$$v(t) = \hat{v}\left(\cos(\omega t)\ \sin(\varphi_v) + \cos(\varphi_v)\ \sin(\omega t)\right). \tag{6}$$

The iscrete Fourier Transform can be defined by the following equations:

$$\begin{aligned} \mathrm{Re}\{v\} &= \frac{2}{N}\sum_{k=n-N+1}^{n}\left|v(k)\ \sin\left(2\pi\ \frac{k}{N}\right)\right| \\ \mathrm{Im}\{v\} &= \frac{2}{N}\sum_{k=n-N+1}^{n}\left|v(k)\ \cos\left(2\pi\ \frac{k}{N}\right)\right| \end{aligned} \tag{7}$$

Applying these equations to the original signal equation results in the following expressions:

$$\begin{aligned} \mathrm{Re}\{V\} &= \hat{v}\ \cos(\varphi_v) \\ \mathrm{Im}\{V\} &= \hat{v}\ \sin(\varphi_v) \end{aligned} \tag{8}$$

The magnitude of the signal phasor can be calculated by the following equation:

$$V = \sqrt{\mathrm{Re}\{V\}^2 + \mathrm{Im}\{V\}^2} \tag{9}$$

The phase angle of the signal phasor can be calculated by the following equation:

$$\varphi = \arctan\left(\frac{\mathrm{Im}\{V\}}{\mathrm{Re}\{V\}}\right) \tag{10}$$

With these definitions, the Discrete Fourier Transform calculation is able to convert the sinusoidal voltage wave shape to a phasor. For each input quantity the measurement will output 3 values:

- the magnitude (also described as instantaneous RMS),
- the real component of the RMS phasor and
- the imaginary component of the RMS phasor.

Fast Fourier Transform
If the number of samples per cycle is great and equal to 2n it could be interesting to use a Fast Fourier Transform (FFT) algorithm instead of the normal DFT method.

Cosine Filter
Another approach as the Fourier transformation is to only use the coefficients from the cosine with the second signal being the cosine value from the calculation 90 electrical degrees previous.

2.3 Transformed quantities outputs

In the case of the three-phase system, the transformed quantities outputs are the real and imaginary parts and magnitude of the positive-, negative- and zero-sequence voltage that are calculated using the symmetrical components transformation. For example of voltage the transformed quantities outputs are calculated as:

$$
\begin{aligned}
U_{0\,\mathrm{r}} + \mathrm{j}\, U_{0\,\mathrm{i}} &= \frac{1}{3} \cdot (\underline{U}_{\mathrm{a}} + \underline{U}_{\mathrm{b}} + \underline{U}_{\mathrm{c}}) \\
U_{1\,\mathrm{r}} + \mathrm{j}\, U_{1\,\mathrm{i}} &= \frac{1}{3} \cdot \left(\underline{U}_{\mathrm{a}} + \left(-\frac{1}{2} + \mathrm{j}\frac{\sqrt{3}}{2} \right) \cdot \underline{U}_{\mathrm{b}} + \left(-\frac{1}{2} - \mathrm{j}\frac{\sqrt{3}}{2} \right) \cdot \underline{U}_{\mathrm{c}} \right) \\
U_{2\,\mathrm{r}} + \mathrm{j}\, U_{2\,\mathrm{i}} &= \frac{1}{3} \cdot \left(\underline{U}_{\mathrm{a}} + \left(-\frac{1}{2} + \mathrm{j}\frac{\sqrt{3}}{2} \right) \cdot \underline{U}_{\mathrm{b}} + \left(-\frac{1}{2} - \mathrm{j}\frac{\sqrt{3}}{2} \right) \cdot \underline{U}_{\mathrm{c}} \right)
\end{aligned}
\tag{11}
$$

The sequence components for current are also calculated with symmetrical components transformation.

3 Measurement Element Input (*RelMeasure*)

The nominal current and the nominal voltage can be selected from the range of settings defined by parameters in the measurement type. The same applies to the current setting (*e:Is*). For the zero sequence value it can be selected between directly measuring or calculate with a check box in the measurement type dialogue or in the measurement element dialogue when it be selected “user defined” in the measurement type dialogue.

4 Calculation Models

For the different number of phases

- Single-phase,
- 3-phases and
- 6-phases

a measurement for current, voltage and their combination is available.

4.1 Single-phase measurement

This measurement type provides for a single phase the current or / and voltage RMS value calculated using the real and imaginary values.

4.1.1 Current

Table 4.1: Parameters of the single-phase current measurement

Parameter	Description	Unit
Inom	Nominal Current	sec.A
Is	Current Setting	p.u.

Table 4.2: Input signals of the single-phase current measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wIr	Current, Real part	sec.A
wIi	Current, Imaginary part	sec.A

Table 4.3: Input signal of the single-phase current measurement for RMS and EMT simulations

Signal	Description	Unit
wIr	Current, Real part	sec.A

Table 4.4: Output signals of the single-phase current measurement

Signal	Description	Unit
I	Current, Magnitude	sec.A
Ir	Current, Real part	sec.A
Ii	Current, Imaginary part	sec.A

4.1.2 Voltage

Table 4.5: Parameter of the single-phase voltage measurement

Parameter	Description	Unit
Unom	Nominal Voltage	sec.V

Table 4.6: Input signals of the single-phase voltage measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wUr	Voltage, Real part	sec.V
wUi	Voltage, Imaginary part	sec.V

Table 4.7: Input signal of the single-phase voltage measurement for RMS and EMT simulations

Signal	Description	Unit
wUr	Voltage, Real part	sec.V

Table 4.8: Output signals of the single-phase voltage measurement

Signal	Description	Unit
U	Voltage, Magnitude	sec.V
Ur	Voltage, Real part	sec.V
Ui	Voltage, Imaginary part	sec.V

4.2 3-phase measurement

This measurement type provides in a 3 phase system the current and / or voltage RMS values for every loop and the sequence components. It can be selected phase-earth and phase-phase loop for current and voltage. The phase-phase values (V_{pp}) are calculated with phase-earth values (V_{pe}) as follows:

$$\begin{aligned} V_{\mathrm{pp\,a}} &= V_{\mathrm{pe\,a}} - V_{\mathrm{pe\,b}} \\ V_{\mathrm{pp\,b}} &= V_{\mathrm{pe\,b}} - V_{\mathrm{pe\,c}} \\ V_{\mathrm{pp\,c}} &= V_{\mathrm{pe\,c}} - V_{\mathrm{pe\,a}} \end{aligned} \tag{12}$$

The current measurement can also calculate the greater between the phase current RMS signals and return this in a separate output signal (“Imax”).

4.2.1 Current

Table 4.9: Parameters of the 3-phase current measurement

Parameter	Description	Unit
Inom	Nominal Current	sec.A
Is	Current Setting	p.u.
calcI0	Calculate I0	
usedLoopCurrent	Current Loop	

Table 4.10: Input signals of the 3-phase current measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wIr_A	Current, Real part A	sec.A
wIi_A	Current, Imaginary part A	sec.A
wIr_B	Current, Real Part B	sec.A

Table 4.10: Input signals of the 3-phase current measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wIi_B	Current, Imaginary part B	sec.A
wIr_C	Current, Real part C	sec.A
wIi_C	Current, Imaginary part C	sec.A
wI0x3r	3*I0, Real part	sec.A
wI0x3i	3*I0, Imaginary part	sec.A

Table 4.11: Input signals of the 3-phase current measurement for RMS and EMT simulations

Signal	Description	Unit
wIr_A	Current, Real part A	sec.A
wIr_B	Current, Real Part B	sec.A
wIr_C	Current, Real part C	sec.A
wI0x3r	3*I0, Real part	sec.A

Table 4.12: Output signals of the 3-phase current measurement

Signal	Description	Unit
I_A	Current, Magnitude A	sec.A
Ir_A	Current, Real part A	sec.A
Ii_A	Current, Imaginary part A	sec.A
I_B	Current, Magnitude B	sec.A
Ir_B	Current, Real part B	sec.A
Ii_B	Current, Imaginary part B	sec.A
I_C	Current, Magnitude C	sec.A
Ir_C	Current, Real part C	sec.A
Ii_C	Current, Imaginary part C	sec.A
Imax	Max. Current, Magnitude	sec.A
I0	Zero Seq. Current, Magnitude	sec.A
I0r	Zero Seq. Current, Real part	sec.A
I0i	Zero Seq. Current, Imaginary part	sec.A
I0x3	3*I0, Magnitude	sec.A
I0x3r	3*I0, Real part	sec.A
I0x3i	3*I0, Imaginary part	sec.A
I1	I1, Magnitude	sec.A
I1r	I1, Real part	sec.A
I1i	I1, Imaginary part	sec.A
I2	I2, Magnitude	sec.A
I2r	I2, Real part	sec.A
I2i	I2, Imaginary part	sec.A
I2x3	3*I2, Magnitude	sec.A
I2x3r	3*I2, Real part	sec.A
I2x3i	3*I2, Imaginary part	sec.A

4.2.2 Voltage

Table 4.13: Parameters of the 3-phase voltage measurement

Parameter	Description	Unit
Unom	Nominal Voltage	sec.V
calcU0	Calculate U0	
usedLoopVoltage	Voltage Loop	

Table 4.14: Input signals of the 3-phase voltage measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wUr_A	Voltage, Real part A	sec.V
wUi_A	Voltage, Imaginary part A	sec.V

Table 4.14: Input signals of the 3-phase voltage measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wUr_B	Voltage, Real Part B	sec.V
wUi_B	Voltage, Imaginary part B	sec.V
wUr_C	Voltage, Real part C	sec.V
wUi_C	Voltage, Imaginary part C	sec.V
wU0x3r	3*U0, Real part	sec.V
wU0x3i	3*U0, Imaginary part	sec.V

Table 4.15: Input signals of the 3-phase voltage measurement for RMS and EMT simulations

Signal	Description	Unit
wUr_A	Voltage, Real part A	sec.V
wUr_B	Voltage, Real Part B	sec.V
wUr_C	Voltage, Real part C	sec.V
wU0x3r	3*U0, Real part	sec.V

Table 4.16: Output signals of the 3-phase voltage measurement

Signal	Description	Unit
U_A	Voltage, Magnitude A	sec.V
Ur_A	Voltage, Real part A	sec.V
Ui_A	Voltage, Imaginary part A	sec.V
U_B	Voltage, Magnitude B	sec.V
Ur_B	Voltage, Real part B	sec.V
Ui_B	Voltage, Imaginary part B	sec.V
U_C	Voltage, Magnitude C	sec.V
Ur_C	Voltage, Real part C	sec.V
Ui_C	Voltage, Imaginary part C	sec.V
U0	Zero Seq. Voltage, Magnitude	sec.V
U0r	Zero Seq. Voltage, Real part	sec.V
U0i	Zero Seq. Voltage, Imaginary part	sec.V
U0x3	3*U0, Magnitude	sec.V
U0x3r	3*U0, Real part	sec.V
U0x3i	3*U0, Imaginary part	sec.V
U1	U1, Magnitude	sec.V
U1r	U1, Real part	sec.V
U1i	U1, Imaginary part	sec.V
U2	U2, Magnitude	sec.V
U2r	U2, Real part	sec.V
U2i	U2, Imaginary part	sec.V
U2x3	3*U2, Magnitude	sec.V
U2x3r	3*U2, Real part	sec.V
U2x3i	3*U2, Imaginary part	sec.V

4.3 6-phase measurement

This measurement type provides in a 3 phases system additionally the phase-phase current and the phase-phase voltage RMS values and instantaneous values (real part and imaginary part). The greater between the phase-earth current RMS signals is returned in a separated output signal(“Imax”). This measurement type provides a whole set of phase-phase current and voltage signals.

4.3.1 Current

Table 4.17: Parameters of the 6-phase current measurement

Parameter	Description	Unit
Inom	Nominal Current	sec.A
Is	Current Setting	p.u.
calcI0	Calculate I0	

Table 4.18: Input signals of the 6-phase current measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wIr_A	Current, Real part A	sec.A
wIi_A	Current, Imaginary part A	sec.A
wIr_B	Current, Real Part B	sec.A
wIi_B	Current, Imaginary part B	sec.A
wIr_C	Current, Real part C	sec.A
wIi_C	Current, Imaginary part C	sec.A
wI0x3r	3*I0, Real part	sec.A
wI0x3i	3*I0, Imaginary part	sec.A

Table 4.19: Input signals of the 6-phase current measurement for RMS and EMT simulations

Signal	Description	Unit
wIr_A	Current, Real part A	sec.A
wIr_B	Current, Real Part B	sec.A
wIr_C	Current, Real part C	sec.A
wI0x3r	3*I0, Real part	sec.A

Table 4.20: Output signals of the 6-phase current measurement

Signal	Description	Unit
I_A	Line-Ground Current, Magnitude A	sec.A
Ir_A	Line-Ground Current, Real part A	sec.A
Ii_A	Line-Ground Current, Imaginary part A	sec.A
I_B	Line-Ground Current, Magnitude B	sec.A
Ir_B	Line-Ground Current, Real part B	sec.A
Ii_B	Line-Ground Current, Imaginary part B	sec.A
I_C	Line-Ground Current, Magnitude C	sec.A
Ir_C	Line-Ground Current, Real part C	sec.A
Ii_C	Line-Ground Current, Imaginary part C	sec.A
Il_A	Line-Line Current, Magnitude A	sec.A
Ilr_A	Line-Line Current, Real part A	sec.A
Ili_A	Line-Line Current, Imaginary part A	sec.A
Il_B	Line-Line Current, Magnitude B	sec.A
Ilr_B	Line-Line Current, Real part B	sec.A
Ili_B	Line-Line Current, Imaginary part B	sec.A
Il_C	Line-Line Current, Magnitude C	sec.A
Ilr_C	Line-Line Current, Real part C	sec.A
Ili_C	Line-Line Current, Imaginary part C	sec.A
Imax	Max. Current, Magnitude	sec.A
I0	Zero Seq. Current, Magnitude	sec.A
I0r	Zero Seq. Current, Real part	sec.A
I0i	Zero Seq. Current, Imaginary part	sec.A
I0x3	3*I0, Magnitude	sec.A
I0x3r	3*I0, Real part	sec.A
I0x3i	3*I0, Imaginary part	sec.A
I1	I1, Magnitude	sec.A
I1r	I1, Real part	sec.A
I1i	I1, Imaginary part	sec.A
I2	I2, Magnitude	sec.A
I2r	I2, Real part	sec.A
I2i	I2, Imaginary part	sec.A

Table 4.20: Output signals of the 6-phase current measurement

Signal	Description	Unit
I2x3	3*I2, Magnitude	sec.A
I2x3r	3*I2, Real part	sec.A
I2x3i	3*I2, Imaginary part	sec.A

4.3.2 Voltage

Table 4.21: Parameters of the 6-phase voltage measurement

Parameter	Description	Unit
Unom	Nominal Voltage	sec.V
calcU0	Calculate U0	

Table 4.22: Input signals of the 6-phase voltage measurement for use in load flow and short-circuit calculations

Signal	Description	Unit
wUr_A	Voltage, Real part A	sec.V
wUi_A	Voltage, Imaginary part A	sec.V
wUr_B	Voltage, Real Part B	sec.V
wUi_B	Voltage, Imaginary part B	sec.V
wUr_C	Voltage, Real part C	sec.V
wUi_C	Voltage, Imaginary part C	sec.V
wU0x3r	3*U0, Real part	sec.V
wU0x3i	3*U0, Imaginary part	sec.V

Table 4.23: Input signals of the 6-phase voltage measurement for RMS and EMT simulations

Signal	Description	Unit
wUr_A	Voltage, Real part A	sec.V
wUr_B	Voltage, Real Part B	sec.V
wUr_C	Voltage, Real part C	sec.V
wU0x3r	3*U0, Real part	sec.V

Table 4.24: Output signals of the 6-phase voltage measurement

Signal	Description	Unit
U_A	Line-Ground Voltage, Magnitude A	sec.V
Ur_A	Line-Ground Voltage, Real part A	sec.V
Ui_A	Line-Ground Voltage, Imaginary part A	sec.V
U_B	Line-Ground Voltage, Magnitude B	sec.V
Ur_B	Line-Ground Voltage, Real part B	sec.V
Ui_B	Line-Ground Voltage, Imaginary part B	sec.V
U_C	Line-Ground Voltage, Magnitude C	sec.V
Ur_C	Line-Ground Voltage, Real part C	sec.V
Ui_C	Line-Ground Voltage, Imaginary part C	sec.V
Ul_A	Line-Line Voltage, Magnitude	sec.V
Ulr_A	Line-Line Voltage, Real part	sec.V
Uli_A	Line-Line Voltage, Imaginary part	sec.V
Ul_B	Line-Line Voltage, Magnitude	sec.V
Ulr_B	Line-Line Voltage, Real part	sec.V
Uli_B	Line-Line Voltage, Imaginary part	sec.V
Ul_C	Line-Line Voltage, Magnitude	sec.V
Ulr_C	Line-Line Voltage, Real part	sec.V
Uli_C	Line-Line Voltage, Imaginary part	sec.V
U0	Zero Seq. Voltage, Magnitude	sec.V
U0r	Zero Seq. Voltage, Real part	sec.V
U0i	Zero Seq. Voltage, Imaginary part	sec.V
U0x3	3*U0, Magnitude	sec.V
U0x3r	3*U0, Real part	sec.V

Table 4.24: Output signals of the 6-phase voltage measurement

Signal	Description	Unit
U0x3i	3*U0, Imaginary part	sec.V
U1	U1, Magnitude	sec.V
U1r	U1, Real part	sec.V
U1i	U1, Imaginary part	sec.V
U2	U2, Magnitude	sec.V
U2r	U2, Real part	sec.V
U2i	U2, Imaginary part	sec.V
U2x3	3*U2, Magnitude	sec.V
U2x3r	3*U2, Real part	sec.V
U2x3i	3*U2, Imaginary part	sec.V

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